

REVISION 2026

Construction Materials Lab

Diploma in Civil Engineering

Code: 2018

Credits: 1.5

Semester: 2

Course Outcomes (CO)

- **CO1:** Identify the physical properties of bricks and prepare cement mortar. (Apply)
- **CO2:** Identify soil types and flooring tiles. (Apply)
- **CO3:** Identify special construction materials and determine bulking of sand. (Apply)

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module 1: Bricks & Mortar

Expt 1: Dimension & Density of Bricks

Measure dimensions of 10 bricks and find average dimension, weight, and density.

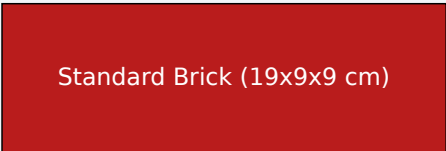
Expt 2: Field Test on Bricks

Perform field tests including dropping, striking, and scratching by nail.

Expt 4: Setting out of Building

Setting out a small building in the field using ropes and pegs.

SVG: Brick Dimension Check



For the purpose of this experiment, a standard brick is used to check the dimensions of the building. The brick is measured in centimeters (cm) and is used to set out the building. The process of setting out is called *Setting out* and is a common practice in construction.

Module 2: Soil & Flooring

Expt 5: Soil Report

Identify various layers and types of soil in the foundation pit and prepare a report.

Expt 6: Flooring Tiles

Identify different types of flooring tiles such as vitrified, ceramic, glazed, and mosaic tiles.

பெரிய அளவுகிலம் கொண்ட பல்வேறு வகைகளில் பிளவுபட்டிருக்கின்றன. இவற்றில் சில பிளவுபட்டிருக்கின்றன. பிளவுபட்டிருக்கின்றன.

Module 3: Special Materials & Sand

Expt 8: Bulking of Sand

Determine the increase in volume of sand due to moisture content (Bulking).

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Practice Model Question Paper

Practice Model Paper — not an official REV2026 paper

1. Find the average weight and density of the given 5 bricks.
2. Prepare a cement mortar of 1:4 proportion for the given quantity of sand.

Source Declaration

Curriculum Scheme: Revision 2026

Subject Name: Construction Materials Lab

Subject Code: 2018

Official Source URL: [SITTTR Link](#)

Source Verification Date: 2026-08-19

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